Milestone II: Model Architecture & Experiments

Due: Module 9–10 (Transformers / Vision Transformers)

Key Components

1. **Abstract:** Concise project summary
2. **Introduction:** Problem context and motivation
3. **Literature Survey:** Expanded from Milestone I
4. **Method:** Detailed architecture description
5. **Preliminary Experiments:** Initial results
6. **Next Steps:** Planned improvements
7. **Member Contributions:** Individual responsibilities

Required Elements:

- At least one baseline model
- Training curves & metrics
- Error analysis
- Ablation study plan

Submission Format:

- Compact, up to 3 pages
- Include code repository URL
- Jupyter notebooks welcome
- `GroupID_checkpoint.pdf`

Experimental Design Guidelines

Baseline Selection:

- Start simple (e.g., ResNet for vision)
- Use pre-trained when applicable
- Document hyperparameters
- Ensure reproducibility

Ablation Studies:

- Vary one component at a time
- Document all changes
- Use consistent seeds
- Track with W&B or TensorBoard

Metrics to Report:

- Task-specific metrics
- Training/validation curves
- Computational efficiency
- Parameter count
- Inference time